# Sharan Sahu

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## Research interests

high-dimensional statistics, stochastic and robust optimization, reinforcement learning, sequential decision-making under distribution shift, and language and diffusion models.

## Education

2024 – Present **Cornell University** – Ithaca, NY
PhD in Statistics and Machine Learning
GPA: 4.103 / 4.000
Advisors: Martin Wells, Yuchen Wu, Y. Samuel Wang

2020 – 2024 **University of California, Berkeley** – Berkeley, CA
BA in Computer Science
GPA: 3.998 / 4.000
Advisors: Iain Carmichael, Ryan Tibshirani

## Publications

* denotes first author(s)

2026 **Provably Reliable Classifier Guidance via Cross-Entropy Control**
Sharan Sahu*, Arisina Banerjee*, Yuchen Wu
*Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS)*

2026 **On the Provable Suboptimality of Momentum SGD in Nonstationary Stochastic Optimization**
Sharan Sahu*, Cameron J. Hogan*, Martin T. Wells
*Forty-Third International Conference on Machine Learning (ICML)*

## Preprints

* denotes first author(s)

2026 **The Price of Current-Effect Inference in Nonstationary Multi-Armed Bandits**
Sharan Sahu*, Martin T. Wells
*Submitted (Annals of Statistics)*

2026 **Adapt or Forget: Provable Tradeoffs Between Adam and SGD in Nonstationary Optimization**
Sharan Sahu*, Abir Sarkar*, Cameron J. Hogan*, Martin T. Wells
*Submitted (ICLR)*

2026 **HOPPER: Learnable Hop Extraction for Linearized Graph Sequence Models**
Isuru Herath*, Arin Gopakumar*, Sharan Sahu*
*Submitted (ICLR)*

2026 **Large Language Model Based Diagnostic Concept Extraction From Gastrointestinal And Breast Pathology Reports**
Dan Qazi, Ayesha Siddika, Sharan Sahu, Michael Superdock, Nabil Rhaoui, Van Hovenga, J Harry Caufield, Michael Sharpnack, Benjamin Calhoun, Kwun Wah Wen, Zoltan Laszik, Iain Carmichael
*Submitted (JAMIA)*

2026 **A General Mixed-Supervision Strategy for Improving Annotation Efficiency in Strongly-Supervised Nuclear Instance Segmentation**
Van Hovenga, Iain Carmichael, Sharan Sahu
*Preprint*

2025 **Online Distributionally Robust LLM Alignment via Regression to Relative Reward**
Sharan Sahu*, Martin T. Wells
*Preprint*

2025 **Towards Optimal Differentially Private Regret Bounds in Linear MDPs**
Sharan Sahu*
*Preprint*

2024 **WSIC: A Python Package To Facilitate Running Nuclear/Cellular Segmentation On Whole Slide Images**
Sharan Sahu*, Jerry Li, Van Hovenga, Kaitlin Smith, Neo Yin, Richard J. Chen, Iain Carmichael
*Preprint*

2023 **Developing Multi-Dimensional Metrics for Precision, Recall, Fidelity, Diversity, and Authenticity in Evaluating Generative Networks Performance using Deep Perceptual Embeddings**
Sharan Sahu*, Daniel Flaherty, Abhishek Vinchure, Jonny Pei, Suya You
*Preprint*

## Invited Talks and Guest Lectures

**Beyond RNNs: An Introduction to Transformers and LLM Basics**
July 2025 Cornell Tech (Break Through Tech AI Program)

**Towards Optimal Differentially Private Regret Bounds in Linear MDPs**
May 2025 Cornell University (Statistics Graduate Society)

**The Machine Learning Problems Behind Large Language Models: Self-Supervision, Fine-Tuning, and Reinforcement Learning**
February 2025 University of North Carolina, Chapel Hill (Computational Pathology Labs)

**Unlocking the Power of Databases: The Crucial Role of Theory and Indices in Scalable Vector Databases for Machine Learning**
August 2024 Naval Postgraduate School (NPS)

**How Do Neural Networks Learn**

December 2023 Naval Postgraduate School (NPS)

## Conference Posters and Oral Presentations

**Provably Reliable Classifier Guidance via Cross-Entropy Control**

August 2026 Joint Statistical Meetings (JSM)

**On the Provable Suboptimality of Momentum SGD in Nonstationary Stochastic Optimization**

August 2026 Joint Statistical Meetings (JSM)

July 2026 International Conference on Machine Learning (ICML)

**DRO-REBEL: Distributionally Robust Relative-Reward Regression for Fast and Efficient LLM Alignment**

September 2025 Cornell University Celebration of Statistics and Data Science

**System Requirements Clustering with Machine Learning and Architecture Design**

February 2025 Naval Applications of Machine Learning (NAML)

March 2024 Naval Applications of Machine Learning (NAML)

March 2023 Neptune Office of Naval Research (ONR) Conference

**Developing Multi-Dimensional Metrics for Precision, Recall, Fidelity, Diversity, and Authenticity in Evaluating Generative Networks Performance using Deep Perceptual Embeddings**

September 2022 DoD 6.1 Research Conference

**Advancing Procedural Scene Synthesis through Enhanced Grammars and Gradient Policies in MetaSim**

September 2021 UC Berkeley Data Science Conference

**Using Machine Learning to Model and Discover New Catalysts To Address The Energy Challenges Posed by Climate Change**

May 2021 UC Berkeley Data Science Conference

## Teaching experience

Fall 2021 **EECS 16A: Foundations of Signals, Dynamical Systems, and Information Processing**

Served as an undergraduate course assistant for the EECS 16AB sequence, delivering foundational instruction in signal processing, control, circuit design, and machine learning with a strong emphasis on practical linear algebra applications. Facilitated both individual and group tutoring sessions, led EECS 16A review workshops, and supported over 50 students in lab settings by connecting theoretical concepts to hands-on implementation. Consistently recognized for teaching excellence, earning an average student rating of 4.9/5.0 for instructional quality, subject expertise, and proactive support.

Spring 2021 **CS 61A: Structure and Interpretation of Computer Programs**
Served as a undergraduate course assistant for the CS 61A series, covering software construction, machine operations, and programming abstraction using Python, Scheme, and SQL. Led weekly tutorials and bi-weekly labs focused on fundamental concepts such as recursion, induction, and data structures including trees and binary search trees. Prepared and delivered review sessions before exams, collaborated on the development of problem sets and assessments, and provided individualized support during office hours, contributing to measurable improvements in student performance.

## Industry experience

Summer 2026 **Microsoft AI**, Research Scientist Intern – Redmond, WA
Safe and Responsible AI agentic systems for Search.

Summer 2023 **US Department of Defense (DoD)**, Data Science Intern – San Diego, CA
System Requirements Automation — Built ML models and full-stack tools for classifying military system requirements, saving $1M and presenting to Pentagon and Marine Corps leadership.

Summer 2022 **US Department of Defense (DoD)**, Software Engineering Intern – San Diego, CA
Defense ML & Cloud Systems — Built ML and cloud tools for EMF waveform classification, automated document parsing, and scalable resource management, boosting detection and efficiency across key pipelines.

Summer 2021 **Novartis**, Product Data Science Intern – Chicago, IL
Operations & Forecasting Analytics — Delivered ML and statistical solutions for inventory forecasting, database optimization, and secure access control, driving $1.5M in savings and boosting system efficiency and security.

2018 - 2020 **Northrop Grumman**, Software Engineering and Data Science Intern – Chicago, IL
Autonomous Systems Research — Developed ML models for sensor fusion and defect detection, combining filtering techniques and CNNs; published findings in a 20-page paper presented at Northrop Grumman's research conference.

## Mentorship and service

2024 – Present **Cornell Statistics Graduate Society**, Vice President
Represented Ph.D. students in academic and departmental affairs, while organizing initiatives to support professional, social, and community development.

**Conference Referee:** NeurIPS; ICML; ICLR

## Skills

**Programming**
Proficient: Python, C++, C, Java
Familiar: R, Rust, C#, Go, Dart

**Software**

LaTeX, Git, Docker, PowerBI, Tableau, Spotfire, Qlik

**Languages**

English (native), Oriya (Advanced)

## Coursework

**Statistics:** High dimensional statistics, asymptotic statistics, mean field asymptotics, mathematical statistics, nonparametric estimation, statistical computing, generalized linear models, empirical process theory, optimal transport, statistical learning theory

**Mathematics:** Real analysis, functional analysis, measure theoretic probability and martingale theory, robust and stochastic optimization, convex and variational analysis, differential manifolds, ring and group theory, random matrix theory, partial differential equations, linear algebra, topology and metric spaces

**Machine Learning:** Machine learning theory (PAC & PAC-Bayes, online convex optimization, stochastic bandits), reinforcement learning theory, deep learning and generative models, recommender and feedback systems, natural language processing, computer vision

**Computer Science:** Data structures and algorithms, operating systems, compilers, computer security, computer architecture, database systems, networking and internet protocols, parallel computing